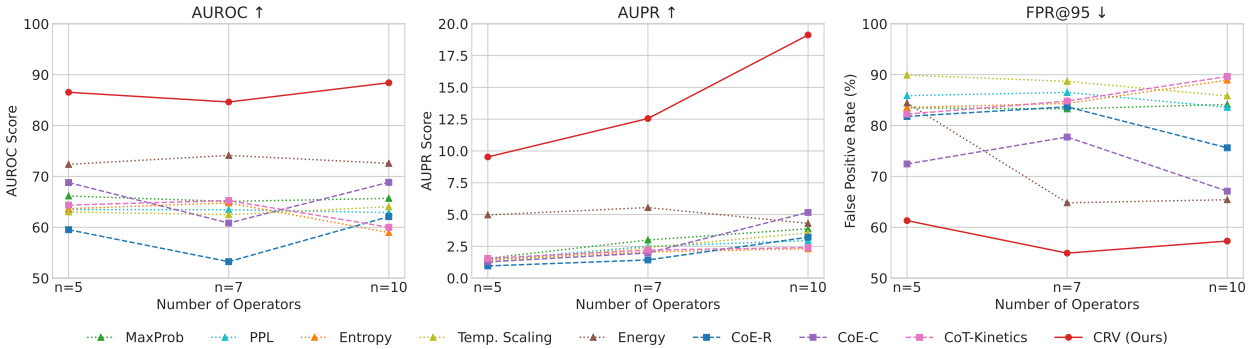


Figure 2 Performance of the step correctness predictors on the synthetic arithmetic task as a function of difficulty (number of operators). CRV retains a clear advantage as complexity increases.

Performance Under Increasing Difficulty. To further probe CRV's robustness, we analyze its performance on the synthetic arithmetic dataset as a function of problem complexity, controlled by the number of operators ($n \in \{5, 7, 10\}$).[5] Figure 2 plots the performance of CRV against key baselines across these difficulty levels. While most methods show stable AUROC and FPR@95, CRV maintains a consistent advantage across all difficulty levels. AUPR generally improves for all methods as difficulty rises because harder problems increase the proportion of incorrect examples (a condition to which AUPR is sensitive). Importantly, CRV's advantage persists despite these shifts, highlighting the robustness of its structural signals across task difficulty and class balance.

4.3 Mechanistic Analysis of Error Computations (RQ2)

Having demonstrated CRV's predictive power, we now turn to its key advantage: interpretability. To address RQ2, we dissect our graph representation to identify structural "fingerprints" of error, from high-level feature ablation to fine-grained analysis of the most predictive structures.

Ablation of Feature Families. A leave-one-out ablation study on the Synthetic (Arithmetic) dataset reveals a clear hierarchy of feature importance, as summarized in Table 3. The *Node Influence & Activation* features are demonstrably the most critical; their removal causes the most performance degradation across all metrics, most notably increasing FPR@95 by over 12 points. The *Global Graph Statistics* also provide a substantial contribution. Interestingly, the *Topological & Path-Based* features appear least critical for this specific task, suggesting that the state of key local features is a more dominant signal than the holistic graph structure. Nevertheless, the full CRV model, which integrates all three signal types, is required to achieve optimal verification performance.

Feature Set	Arithmetic		
	AUROC ↑	AUPR ↑	FPR@95 ↓
CRV (All three families)	**92.47**	**28.92**	**37.09**
Ablation:			
– w/o Global Stats	89.62	24.35	44.54
– w/o Node Stats	88.31	23.25	49.07
– w/o Topological Stats	90.89	26.83	39.19

Table 3 Leave-one-out ablation study on the Synthetic (Arithmetic) dataset.

Visualizing the Structural Signatures of Error. To provide qualitative evidence for our hypothesis, we visualize the "structural fingerprints" learned by our classifier. Figure 4 shows distributions of five highly predictive features for correct versus incorrect GSM8K reasoning steps. Across diverse feature types, from graph topology (e.g., *Graph Density*) to node statistics (e.g., *Total Active Features*), distributions are clearly distinct. Similar patterns are observed on our synthetic datasets (see included in Appendix C.5), confirming that the graph representation captures separable structural differences between valid and flawed computations.

While individual features are predictive, CRV's strength lies in their combination. To illustrate this, we project the full high-dimensional feature vectors into two dimensions via Principal Component Analysis (PCA). Figure 3 reveals that incorrect steps form a dense subset *within* the broader distribution of correct steps.

[5] We exclude $n = 3$ as the model's high accuracy yields too few incorrect examples for reliable evaluation.

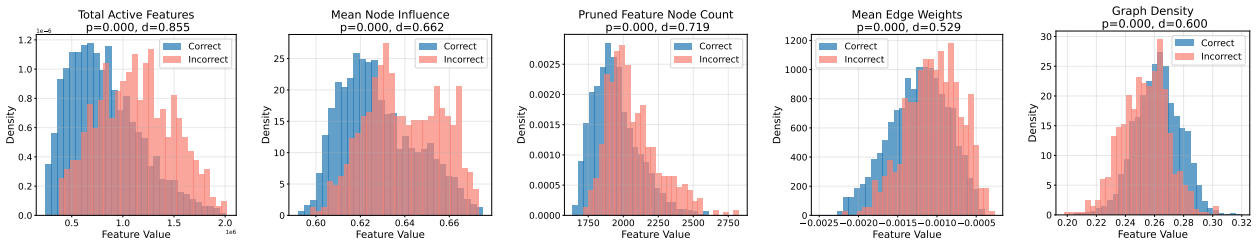
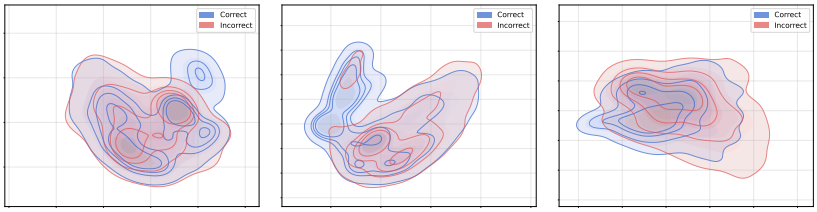


Figure 4 Topological Fingerprints of Error on GSM8K. Distributions of five selected graph features for correct (blue) vs. incorrect (red) reasoning steps. The visual separation is statistically significant for each feature shown (independent t-test, $p < 0.001$) and represents a medium-to-large effect size (Cohen's d). This provides quantitative evidence that attribution graphs contain a clear, separable structural signal of a computation's integrity.

Crucially, correct steps also occupy a distinct region not shared by incorrect computations. This suggests many reasoning failures are computational "near misses", structurally similar to valid steps, which explains the overlap. Yet there exists a *zone of computational integrity*, a region defined by structural properties accessible only to correct reasoning. A complementary t-SNE visualization in Appendix C.5 shows the same structure, confirming that CRV succeeds by learning the complex boundary of this zone rather than separating two simple clusters.



(a) Boolean **(b)** Arithmetic **(c)** GSM8K

Figure 3 Distributions of features after PCA for correct (blue) vs. incorrect (red) reasoning steps.

4.4 From Diagnosis to Causal Intervention (RQ3)

Finally, we conduct an exploratory study to test whether CRV's diagnostic insights can guide prescriptive interventions. A key advantage of our white-box approach is that failure-predictive graph features can be traced to specific components of the model's computation. We illustrate this with a case study on the arithmetic task, correcting an order-of-operations error.

Given the expression `(7*((5+9)+7))`, the model incorrectly computed `7*14=98` in its second step (Table 4). CRV flagged this step as incorrect, and feature importance highlighted an unusually high activation of a late-layer transcoder feature. Tracing the signal back to the graph, we found a single highly active last-layer transcoder feature (ID 91814) strongly associated with multiplication.

We hypothesized that this premature activation caused the error, and performed a direct intervention. We re-ran the generation up to the point of failure and used a forward hook to manually clamp the activation of this specific *multiplication* feature to zero. The effect was immediate: as shown in the right column of Table 4, with the premature multiply impulse suppressed, the model correctly generated the next step `14+7=21` and proceeded to the correct final answer.

While not a general-purpose solution, this proof-of-concept is a crucial step. Beyond suppressing faulty features, we also corrected an error by *amplifying* an under-active feature (Appendix C.5). The success of both interventions provides closed-loop evidence that CRV's structural signatures are causally implicated in errors, opening a promising direction for targeted model interventions. Crucially, because these interventions rely on the semantic interpretation of specific transcoder features (e.g., identifying a missing "multiplication" concept), they demonstrate the potential of CRV to serve as a foundation for interpretable debugging, moving beyond simple error flagging to actionable model repair.

Before Intervention (Incorrect)	After Intervention (Correct)
Evaluate the arithmetic expression below. (7 * ((5 + 9) + 7)) To evaluate this expression, we need to follow the order of operations (PEMDAS): 1. Evaluate the expression inside the innermost parentheses: 5 + 9 = 14 2. Multiply 7 by the result: 7 * 14 = 98 3. Add 7 to the result: 98 + 7 = 105 Therefore, the expression evaluates to 105.	Evaluate the arithmetic expression below. (7 * ((5 + 9) + 7)) To evaluate this expression, we need to follow the order of operations (PEMDAS): 1. Evaluate the expression inside the innermost parentheses: 5 + 9 = 14 2. Add 7 to the result: 14 + 7 = 21 3. Multiply 7 by the result 7 * 21 = 147 Therefore, the value of the expression is 147.

Table 4 Side-by-side comparison of a reasoning trace before and after causal intervention. The highlight indicates the point of divergence where suppressing a single *multiplication* transcoder feature corrects the model's computational path.

5 Related Work

CoT Prompting for Improved Reasoning. Chain-of-Thought (CoT) prompting was introduced by Wei et al. (2022) to elicit more complex reasoning from LLMs through few-shot examples that demonstrate step-by-step problem solving. Subsequent work has expanded this idea in several directions. Kojima et al. (2022) showed that even a simple zero-shot instruction such as "Let's think step by step" can trigger coherent reasoning traces. While this reduces the need for handcrafted prompts, providing structured examples often remains beneficial. To scale this process, recent studies generate CoT exemplars synthetically (Zhang et al., 2023; Shao et al., 2023; Li et al., 2025a). Other work leverages test-time compute scaling to extend reasoning chains, enabling longer and more elaborate solutions (Snell et al., 2024). For comprehensive surveys of CoT techniques and their applications, see Chu et al. (2024) and Chen et al. (2025a).

Verifying and Improving CoT Reasoning. The transparency of CoT has also made it a focal point for research into model interpretability and reliability. While some work assumes reasoning traces are to some extent faithful representations of the model's internal process (Wei Jie et al., 2024; Korbak et al., 2025), a significant body of evidence highlights their unreliability (Arcuschin et al., 2025; Bentham et al., 2024; Chen et al., 2025b; Turpin et al., 2023). This has spurred a field of research dedicated to verifying and improving CoT traces. This research broadly investigates (i) the model's intrinsic ability to self-evaluate its reasoning steps (Zhang et al., 2025), (ii) how to measure the faithfulness of a reasoning chain to the final answer (Lanham et al., 2023; Bi et al., 2025; Tutek et al., 2025), and (iii) when reasoning steps are needed or useful (Bogdan et al., 2025; Wang et al., 2025b). A parallel line of work aims to improve reasoning chains through various forms of neuro-symbolic reasoning (Lyu et al., 2023), correction (Tyen et al., 2024), uncertainty calibration (Ji et al., 2025), or by enforcing internal consistency (Xie et al., 2025; Wang et al., 2025a). A distinct approach involves training auxiliary models, such as Process Reward Models (PRMs), to assess step-level correctness and guide post-training (Lightman et al., 2024; Wang et al., 2024; Guan et al., 2025). While all these methods aim to improve reasoning outcomes, they primarily operate on the textual or hidden state representations. We are not aware of previous attempts to verify reasoning by analyzing the structural properties of its underlying computational graph.

Mechanistic Interpretability of CoT Reasoning. Our work is most directly situated within the field of mechanistic interpretability, which seeks to reverse-engineer the algorithms learned by neural networks, moving beyond the surface-level analysis of CoT traces (Wei Jie et al., 2024; Korbak et al., 2025; Baker et al., 2025). A central tenet of this field is that models develop specialized subgraphs, or circuits, to perform specific computations (Olah et al., 2020). Recent work has begun to apply this lens to reasoning, providing evidence that CoT traces are driven by latent learned algorithms (Venhoff et al., 2025) and even characterizing specific self-verification circuits in toy models (Lee et al., 2025). Other research aims to improve performance by eliciting or steering behavioral circuits (Zhao et al., 2025; Ward et al., 2025). A particularly powerful and increasingly popular tool in this area is the use of sparse autoencoders (SAEs), which learn to decompose a model's dense activation vectors into a sparse basis of interpretable features (Bricken et al., 2023; Cunningham et al., 2023). Our work builds directly on a variant, the transcoder (Dunefsky et al., 2025), which acts as a functional, interpretable substitute for an MLP module. While prior work has used transcoder-based attribution graphs to qualitatively analyze the faithfulness of CoT reasoning (Ameisen et al., 2025), our work is the first to operationalize this approach for automated verification. We move beyond visual inspection by systematically extracting quantitative, structural features from these graphs and demonstrating that they can be used to diagnose computational failures.